Yanis Bendehane

Engineering Student in Computer Science | Apprentice Software Engineer at Thales

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Summary

Computer science engineering student in an apprenticeship at Thales, developing mission-critical defense software in Java. Interested in software architecture, legacy modernization and software quality. Available for a 17-week research internship abroad, mid-June to mid-October 2027.

Education

CESI School of Engineering, MEng in Computer Science – Aix-en-Provence, France Sept 2023 – Sept 2028

- French Diplôme d'Ingénieur (CTI-accredited, Master's level): integrated preparatory cycle 2023–2025, apprenticeship engineering cycle with Thales 2025–2028
- Engineering cycle: semester 5 fully validated with grade A in every teaching unit; operations research unit graded A

Lycée Jean Monnet, Bac in Mathematics and Physics-Chemistry, with honors (Mention Bien) – Vitrolles, France Sept 2020 – July 2023

Experience

Apprentice Software Engineer, Thales Services Numériques – Aix-en-Provence, France Oct 2025 – present

- Develop and test features of a large-scale, mission-critical defense software system in Java on Linux, within a 40-person engineering team spread over three sites
- Implemented the display and gateway layers of a software evolution in an MVC-style architecture, from specification to integration testing
- Ramped up on a legacy codebase built on proprietary frameworks with little documentation, through code reading and knowledge transfer from senior engineers

IT Intern, LyondellBasell – Berre-l'Étang, France Apr 2025 – June 2025

- Rebuilt a legacy C++ (MFC) program as a standalone C# Windows service that synchronizes tank logistics states between a real-time industrial database (AspenTech IP.21) and SQL Server over ODBC

Academic Projects

Backup software in C# .NET 8 (team of 4, Software Engineering module) Jan 2026 – Feb 2026

- Delivered three versions: an MVC console app, an Avalonia desktop interface in MVVM with file encryption, then multithreaded backup jobs with priorities and centralized logging
- Full and differential backups, real-time progress tracking, two interface languages; logging packaged as a reusable DLL (JSON/XML), Git branching workflow

Microservices social network (team of 3, Distributed Applications module) June 2026

- Owned the authentication service (Node.js, Express, PostgreSQL, Sequelize, JWT with refresh tokens, bcrypt) and the infrastructure layer
- Deployed 7 Docker containers with Docker Compose behind an Nginx API gateway validating JWTs before routing

Vehicle routing with time windows (team of 4, Operations Research module) Spring 2026

- Modeled an NP-hard VRPTW (capacity, time windows, horizon) and benchmarked four methods on 150 generated instances (10 to 200 customers) against an OR-Tools reference
- Implemented nearest-neighbor + 2-opt, simulated annealing, a genetic algorithm and a Transformer attention model trained with REINFORCE: at 200 customers it stayed 40% from the reference in under 1 s, versus 156% for simulated annealing

Leadership and Volunteering

Co-organizer, charity football tournament (CESI civic engineering project) Jan 2024 – June 2026

- Organized three editions with a team of five; proceeds donated to charities chosen by the winning teams (incl. Les Restos du Cœur, ShareTheMeal)

Skills

Programming: Java, C#, C++, C, Python, JavaScript, TypeScript, SQL

Engineering: OOP, MVC and MVVM, UML and SysML, legacy code maintenance, testing, documentation

Tools: Linux, Git, Subversion, Docker, IntelliJ IDEA, Visual Studio, .NET 8

Languages: French (native), English (B2, TOEIC practice test 865/990), Spanish (school level), Arabic (basic)

Certification: ANSSI SecNumacadémie (French national cybersecurity MOOC)